JAINAM PATEL

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Summary

Entry-level AI Engineer with experience building end-to-end analytics and AI solutions. Proficient in Python, SQL, Spark, LLMs, prompt engineering, and RAG frameworks. Developed predictive models and knowledge graphs and deployed interactive dashboards to translate complex data into business insights. Eager to prototype and deliver agentic AI tools that streamline operations.

Education

McMaster University — Jan 2024 - Dec 2025
Master of Engineering, Systems and Technology | GPA: 3.8/4.0 — *Hamilton, ON*

Gujarat Technological University — Jul 2019 - May 2023
Bachelor of Engineering, Computer Science and Engineering | GPA: 3.6/4.0 — *Gujarat, IN*

Experience

Hydro One Networks Inc. — Jan 2025 - Aug 2025
Data Scientist — *Toronto, ON*

- Built predictive models in Python and scikit-learn to forecast vulnerability trends across 1,000+ assets and used LLM-based NER to build knowledge graphs enabling risk-weighted IT budget decisions.
- Designed and validated data quality testing frameworks on large structured and unstructured datasets using Spark and data blending techniques; built comprehensive Power BI and Matplotlib dashboards to surface vulnerability trends, patch compliance rates, and risk exposure metrics, translating complex data signals into actionable insights for executive stakeholders.
- Applied mathematical optimization techniques to tune forecasting model hyperparameters, reducing prediction error by 15% and improving IT risk quantification accuracy.
- Developed proof-of-concept deep learning models to identify patterns in cybersecurity logs, improving threat detection capabilities by 18%.
- Automated MySQL upgrade orchestration across 500+ systems and PowerShell-based patching across 1,000+ devices, using data-driven methods to flag high-risk assets and improve IT risk forecasting.

Projects

RAGBench: Multi-Architecture RAG Benchmarking Framework — May 2026 - Jun 2026

- Implemented a benchmarking framework in Python comparing 8 RAG architectures (Naive, Self-RAG, CRAG, Graph, Agentic, Modular, Multimodal, Adaptive) using LangChain, LlamaIndex, GPT-4o mini, ChromaDB, and FAISS.
- Developed LLM-as-Judge evaluation scoring faithfulness, relevance, and context precision across 12 queries; Agentic RAG achieved 20% higher faithfulness than Naive RAG with 3x latency trade-off across 8 variants.

Mars Rover Data Pipeline — Jan 2026 - Mar 2026

- Architected and implemented an automated ETL pipeline with Python and Apache Airflow to ingest NASA API data, perform OpenCV feature extraction, and load results into PostgreSQL with conditional branching and retry logic to ensure data integrity.
- Scheduled and orchestrated multi-step workflows mirroring LangGraph-style node-edge execution graphs to analyze image properties, including RGB channels, brightness levels, aspect ratios, and dimensional metadata.

UCI Adult Income Prediction — Jan 2025 - Mar 2025

- Developed and validated Logistic Regression, Decision Tree, and SVM models using scikit-learn, achieving 85% accuracy and 0.91 AUC through stratified cross-validation and hyperparameter tuning.
- Built a robust preprocessing pipeline, handling missing values, encoding categoricals, scaling features, and adding unit tests to ensure data integrity.

Technical Skills

- **Programming & Data Analysis:** Python, SQL, Statistical Analysis, Power BI, R, Scikit-learn, Data Science
- **Databases:** PostgreSQL, Oracle, DB2, MySQL, MongoDB, Redis, DynamoDB
- **Generative AI:** LLMs, Prompt Engineering, RAG, Hugging Face Transformers, Vector Databases, LangGraph, LlamaIndex, Azure OpenAI, AI Copilot, NLP, Knowledge Graphs, Fine-Tuning, Agent Frameworks
- **Machine Learning:** Linear/Logistic Regression, SVM, K-Means, Random Forest, LSTM, CNN, RNN, XGBoost, PyTorch, MLOps, Mathematical Optimization, Deep Learning
- **Cloud Services:** Azure ML, Azure OpenAI, Azure AI Services, Microsoft Fabric, AWS (EC2, S3, Lambda, RDS)